Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Ceylon InnovAI (Maniesha Abeysekera, Ashen Weerasinghe)**
b. Team leader name: **Maniesha Abeysekera**
c. Problem Statement: **Manufacturing: Intelligent Operations & Industrial Efficiency**

Brief about the idea

SiteGuard AI turns a phone photo and a voice note into a verified, auditable safety record.

On factory floors, warehouses and industrial sites, hazards such as blocked walkways, open edges and missing PPE are still logged on paper — inconsistent between inspectors, hard to track to a fix, and almost never verified once "fixed".

1. **Capture**
   A reviewer photographs the hazard and records a voice note.

2. **Gemini proposes**
   One multimodal Gemini call returns a structured finding with a calibrated confidence score.

3. **People decide**
   A human confirms, assigns and closes every record; Gemini verifies the fix against the original evidence.



Real evidence: stock and cardboard spilling into a warehouse walkway (Aisle 3).

Theme: Manufacturing — Intelligent Operations & Industrial Efficiency. Operational evidence becomes findings, actions and insights.

Opportunities

1

How is it different from existing ideas?

- Most safety tools are digital checklists, or vision models that output a verdict.
- SiteGuard reads photo + voice + context in one Gemini call, says "inconclusive" when evidence is weak, and verifies closure by comparing before/after evidence for that specific hazard.
- The AI never decides — a person approves every state change.

2

How will it solve the problem?

- Replaces paper binders with one consistent, structured finding per photo.
- Every hazard gets an owner, a due date and an escalation rule.
- An append-only audit trail links each decision to its evidence and reviewer.
- Site Insights shows managers hotspots, repeat hazards and overdue actions.

3

USP of the proposed solution

- Every hazard seen. Every fix verified.
- AI-verified closure with a human in the loop — not just detection.
- Numbers are computed, not generated: risk and insight counts come from deterministic Python.
- A new hazard class is a new prompt, not new infrastructure.

List of features offered by the solution

1 Multimodal capture

Photo, voice note, typed note, location and time — analysed together in one Gemini call.

2 Structured AI finding

Category, impact, suggested actions and owner team, enforced by a response schema.

3 Calibrated confidence

Confidence score plus an explicit "inconclusive" when angle or lighting prevent a clear read.

4 Deterministic risk matrix

Likelihood × severity scored in Python, so every risk level is defensible.

5 Actions & escalation

Owner, priority, target date and escalation rule for every corrective action.

6 Audit trail & alerts

Append-only history and a notification feed built from it; HTML reports and CSV export.

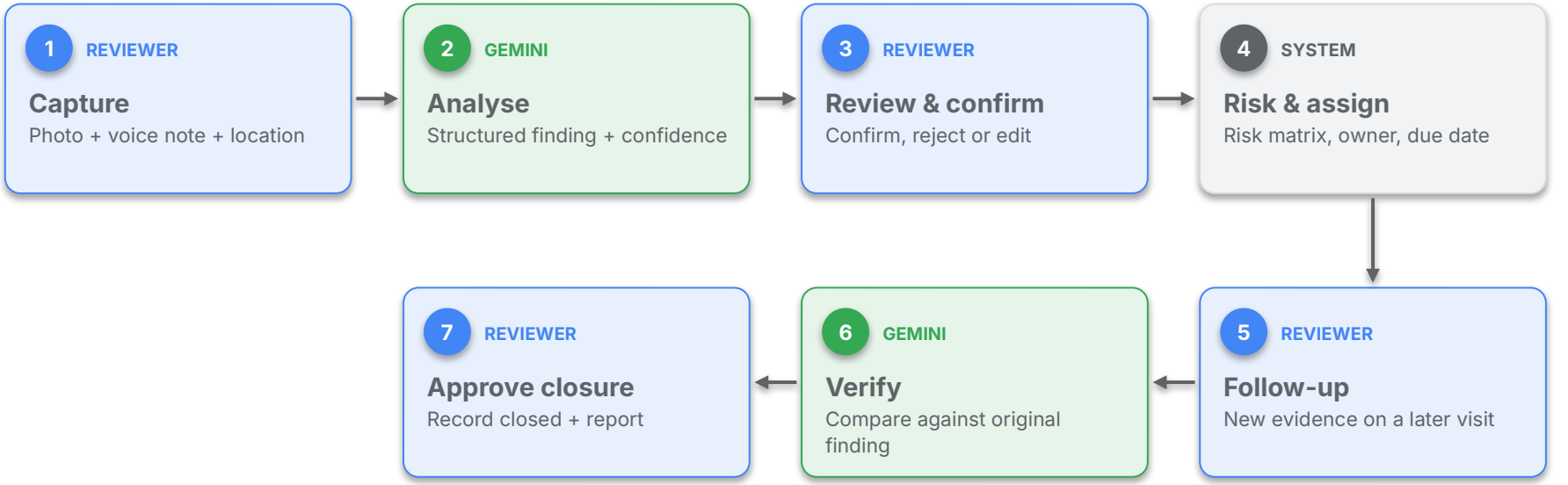
7 AI closure verification

Gemini compares new evidence with the original finding; the reviewer approves or rejects.

8 Site Insights

Gemini briefs managers on recurring hazards, hotspots, overdue teams and blind spots.

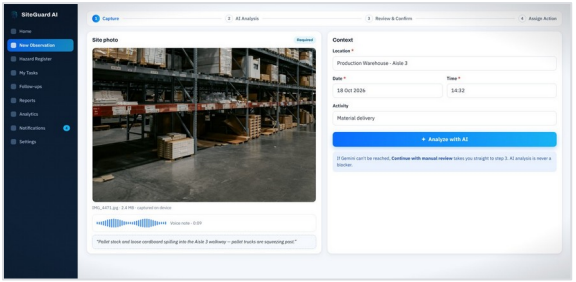
Process flow diagram



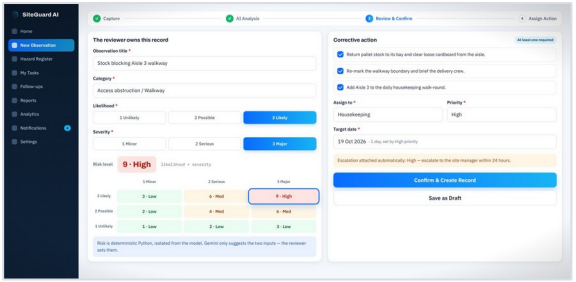
★ **Site Insights (manager)** Python computes the register digest → Gemini interprets it → focus areas, repeat hazards, overdue teams and blind spots.

Every state change in steps 1–7 is written to the append-only audit trail and feeds the notification feed.

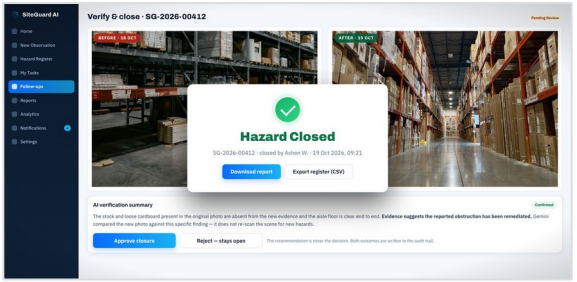
Wireframes / Mock diagrams of the proposed solution



1 · Capture: photo, voice note, context



2 · Review & confirm: risk matrix and actions

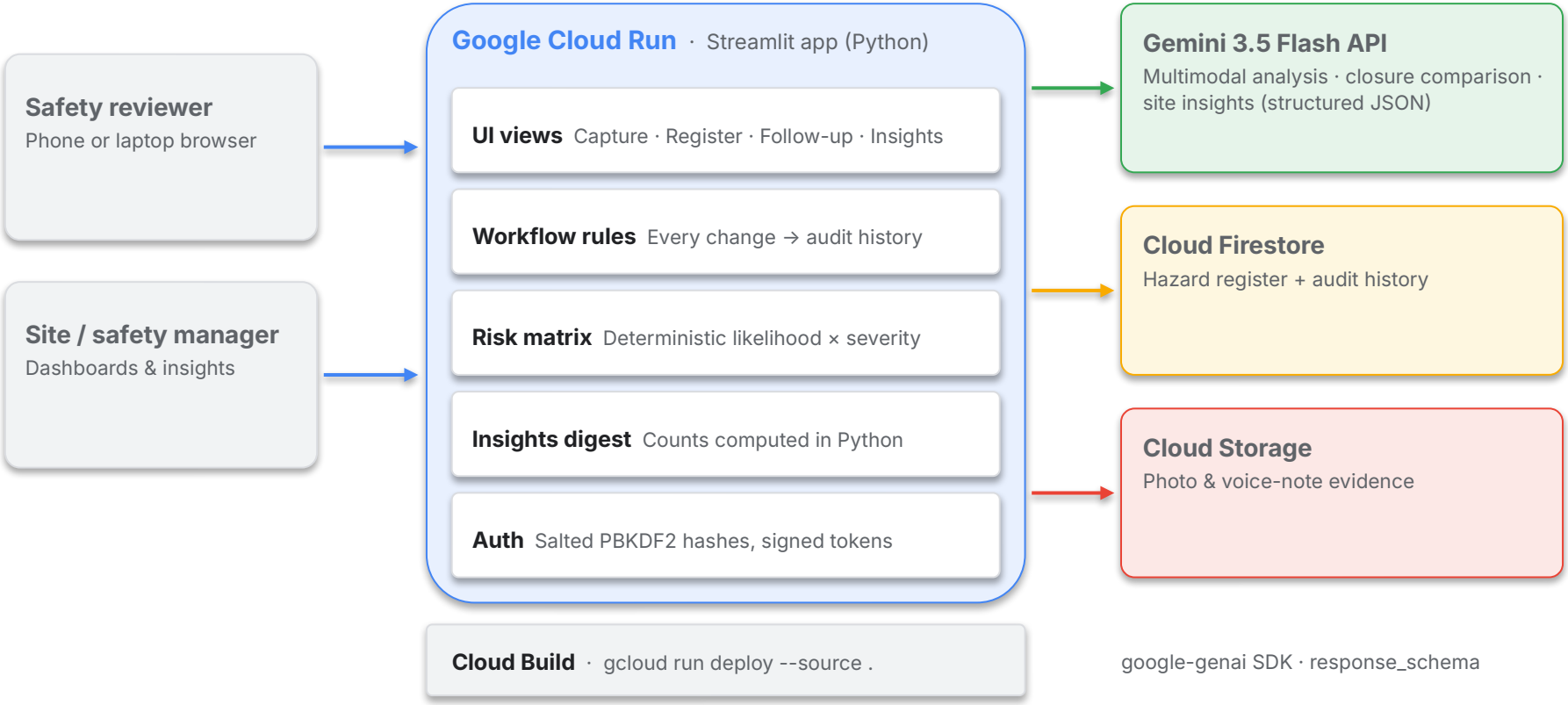


3 · Verify & close: before/after evidence

Design principles

- Mobile-first capture in under a minute
- AI output is always labelled a proposal
- Mandatory fields marked and validated
- Status colours: Open, In Progress, Closed

Architecture diagram of the proposed solution



Technologies to be used in the solution

Layer	Technology	Used for
Generative AI	Gemini 3.5 Flash via the google-genai SDK	Multimodal hazard analysis, before/after closure comparison, register-wide Site Insights
AI output control	response_schema + Pydantic models	Schema-enforced JSON: category, confidence, risk inputs, actions
Hosting	Google Cloud Run (container, scales to zero)	Public, serverless deployment of the app
CI / build	Google Cloud Build	One-command deploy from source
Database	Google Cloud Firestore (Native mode)	Hazard register and append-only audit history
Object storage	Google Cloud Storage	Photo and voice-note evidence linked to each finding
Application	Python 3, Streamlit, pandas	UI, workflow rules, deterministic risk matrix, analytics, CSV/HTML reports
Security	Salted PBKDF2-SHA256, signed session tokens, lockout	Every action attributed to a signed-in reviewer

Estimated implementation cost (optional)

Prototype today

$0

Runs end to end inside Google Cloud free tiers: Cloud Run, Cloud Build, Firestore, Cloud Storage and the Gemini API free tier.

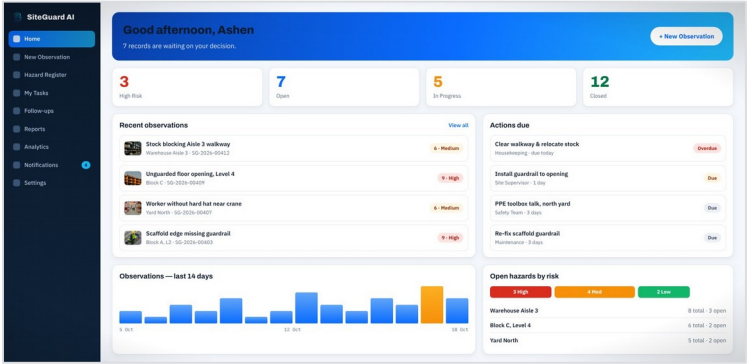
Cloud Run scales to zero, so an idle site costs nothing.

How cost scales in production (usage-based)

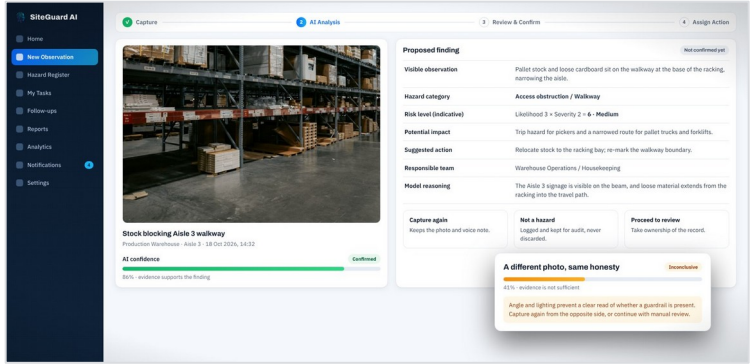
Gemini API	Per token: 1 call per observation, 1 per follow-up, 1 per Site Insights briefing. Photos are the main input; text output is small.
Cloud Run	Per request-second while the app is serving; zero when idle.
Firestore	Per document read/write; each hazard is one small document plus history entries.
Cloud Storage	Per GB stored; one photo (and optional voice note) per observation or follow-up.

Site estimate = observations per day × these drivers, at published Google Cloud rates.

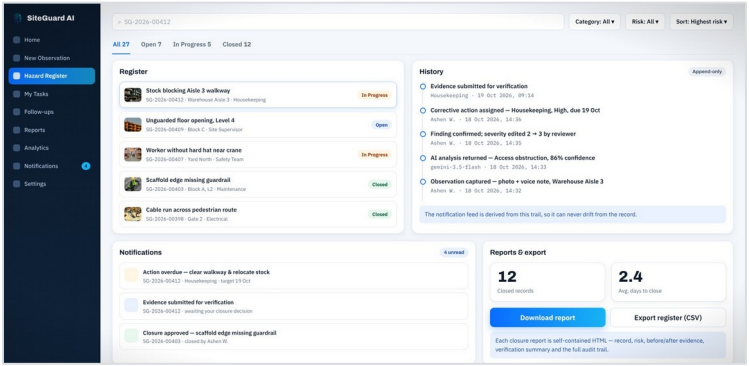
Snapshots of the prototype



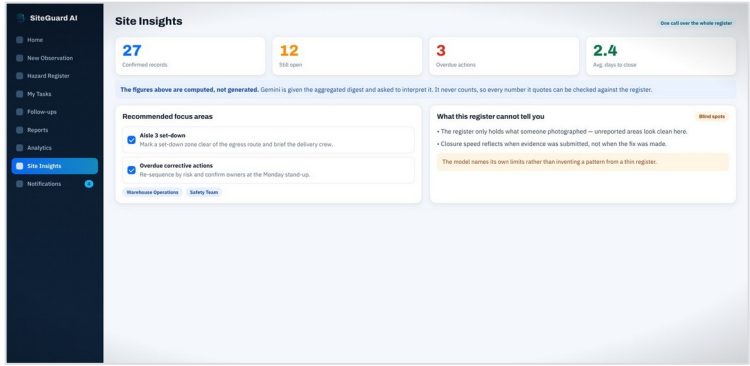
Dashboard: counts, recent observations, actions due



Gemini proposal with confidence and an "inconclusive" example



Hazard register with append-only history



Site Insights: computed counts + Gemini briefing

Prototype Performance report / Benchmarking

1

Gemini call per observation (photo + voice + context together)

0

regex parsing — output is schema-validated JSON

4×

automatic retries on Gemini 503 (2 s, 5 s, 10 s back-off)

100%

state changes need a human and are written to the audit trail

Evaluation set

- 9 real site photos across 3 hazard classes: unprotected edge, obstructed walkway, PPE concern.
- Each class has clear and deliberately ambiguous images, plus "after" photos for closure.
- Expected: clear → correct category; ambiguous → "uncertain"; after → closure supported.

Reliability by design

- Risk is deterministic: the same likelihood and severity always give the same score.
- If Gemini is unreachable, "Continue with manual review" keeps the workflow moving.
- Site Insights needs at least 5 confirmed records before it reads patterns, and names its blind spots.
- Offline demo mode with clearly labelled mock AI for unreliable networks.

Additional Details / Future Development

1 More hazard classes

Lockout-tagout, blocked fire exits, spills, machine guarding — each a new prompt and schema value.

2 Local languages

Voice notes in Sinhala, Tamil and other languages, using Gemini's multilingual audio understanding.

3 Plant-wide benchmarking

Compare sites and shifts; trend repeat hazards and time-to-close across a whole operation.

4 Enterprise integration

Push corrective actions to CMMS/ERP tools; SSO and signed evidence URLs for production.

5 Offline mobile capture

Capture on the floor without signal; sync and analyse when the device reconnects.

6 Evaluation pipeline

Track accuracy and calibration on a growing labelled set with every prompt change.

Provide links to your:

- GitHub Public Repository: https://github.com/Ashenweerasinghe/siteguard-ai
- Demo Video Link (3 Minutes): https://www.youtube.com/watch?v=BarPCIAyHYA
- Final Product Link: https://siteguard-ai-463547683061.us-central1.run.app/

Demo login for judges

Username: **demo** Password: **Safeguard@123** (Safety Officer role)

Team Ceylon InnovAI · Maniesha Abeysekera (team leader) · Ashen Weerasinghe

Responsible AI by design

1 A human approves every state change

Confirm, assign and close all need a reviewer's click, attributed by name and time.

2 No verdicts — only evidence

Gemini never says "safe", "compliant" or "fixed"; only "evidence suggests" or "inconclusive".

3 Every finding links to its source

Each record keeps the original photo and voice note, so any finding can be checked.

4 Computed, not generated

Risk scores and insight counts come from Python; Gemini interprets, it never counts.

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Thank you

